

Visual Attention and Virtual Human Facial Animations in Virtual Reality (VR): An Eye-Tracking Study

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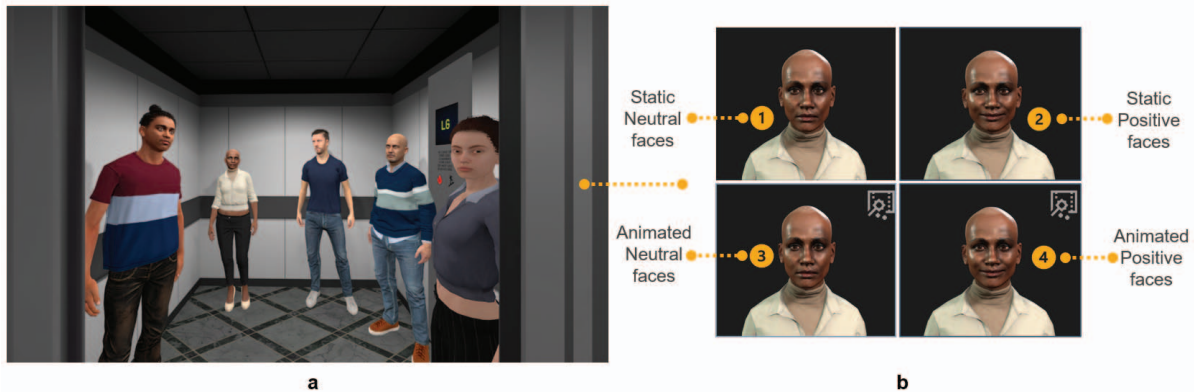
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Figure 1: (a) VR lift environment: view of the lift when the door opened with five virtual humans standing inside. (b) Factorial design with virtual human faces varied in animation (static or animated) and expressions (neutral or positive).

ABSTRACT

We set out to understand the visual attention and perception of virtual humans in individuals vulnerable to paranoia in a VR eye-tracking study. In a factorial between-groups design, 122 individuals with elevated paranoia experienced a virtual lift ride with virtual humans that varied in facial animation (static or animated) and expression (neutral or positive). Facial animation ($p = 0.053$) led to a significant reduction in co-presence. Positive expressions ($p\text{-adj} = 0.046$) significantly decreased the visual attention to virtual humans when their faces were static. Our results indicate that virtual human programming could influence user perception and visual behaviours for people with mistrust.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Empirical studies in HCI

1 INTRODUCTION

Facial expressions on virtual humans could impact people's behavioural and psychological responses in virtual reality (VR), and it has been used to simulate social interaction for mental health insights. Geraets et al. [3] demonstrated that the emotion of virtual human faces affected people's emotion recognition ability that surprise and anger were recognized more accurately than disgust and happiness. Visual attention also focused more on virtual humans' eye and nose areas. Wei et al. [7] found that positive facial expressions on virtual coaches improved user engagement and their connection with the coach in VR mental health consultation.

Paranoia, characterised as perceiving harmful intent where there is

none, has also been studied in VR by examining reactions to virtual humans. It is suggested that closely observing virtual humans might lead to paranoid thoughts [1]. Yet, the specific influence of facial expressions on visual attention and paranoia in VR has not been investigated. Existing research has used eye-tracking to study facial expressions in tasks like threat detection or emotion recognition [3]. The more natural, spontaneous responses to contextual facial expressions are under-explored.

This study investigates the impact of the programming of virtual human faces on people with elevated paranoia. We focused on two aspects of implementing facial expressions: whether their faces are animated and their expression. Through the eye-tracking-enabled VR experience, we looked at people's visual attention, co-presence, and thoughts towards virtual humans (e.g. paranoid appraisals). The study results will contribute to the understanding and decision-making of virtual human design to make VR scenarios in mental health therapies more effective.

2 STUDY DESIGN

We used a two-by-two factor, between-groups, randomised design. The independent variables are **Animation** (static or animated) and **Expression** (neutral or positive) of virtual human faces. 122 participants with elevated paranoid tendencies (screened by *The Revised Green et al., Paranoid Thoughts Scale (R-GPTS)* [2]) experienced a VR lift ride with virtual humans under one of the four facial conditions (static neutral, animated neutral, static positive, animated positive). We recorded participants' eye gaze behaviour in VR. Post-VR, we assessed participants' co-presence [4] and their thoughts about virtual humans.

2.1 VR Experience

We implemented a VR lift ride experience. Participants were instructed to take the lift to the "Sky Lounge" destination. During the three-minute-long lift ride, five other virtual humans (3 males, 2 females) are also standing around (see Fig. 1-a). The VR scenario was developed in Unity 2021.3.15 with Oculus' Movement SDK

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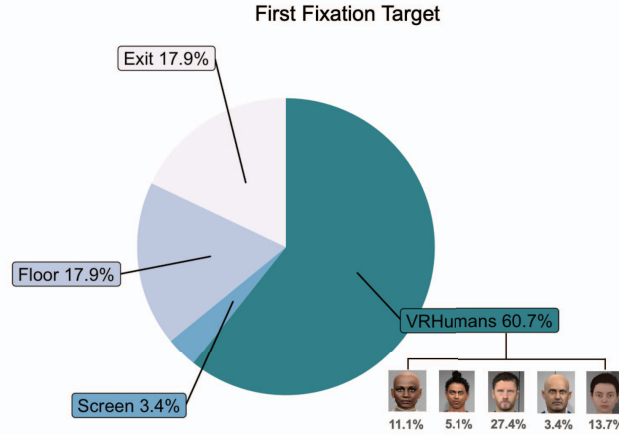


Figure 2: Distribution of the target of the first fixation.

V1.3.2. The experience was rendered on a desktop Windows PC and sent to a Meta Quest Pro via Meta Air Link.

2.2 Animation Creation

We combined motion capture and blend shape animation to create realistic facial animations on virtual humans. We first motion-captured the facial movements as the base motion, and then refined them with keyframe blend shapes for smoother animation. The animations were recorded and processed using Iclone7 and LiveFace plugin¹. Additionally, we implemented an intermittent "look-at" effect, allowing the virtual humans to make eye contact with participants. All virtual humans were given idle body animations.

2.3 Eye-tracking Configuration for Meta Quest Pro

We predefined the regions of interest (ROIs) in VR for eye gaze recording. These ROIs include the faces of virtual humans and specific areas within the lift, such as the floor, ceiling, screen, and exit, to capture the primary focus of visual attention.

Participants' gaze points were calculated in real-time using the Unity built-in ray casting function to determine if gaze rays intersected with ROI objects. We adjusted the scale of scene objects and colliders to account for tracking errors identified in our accuracy test of the Meta Quest Pro's eye tracker (accuracy: 1.652°; SD precision: 1.652°) [6]. Considering our design of the life-scaled virtual humans and faces (width range: 0.13-0.15m), we established a distance threshold of under 3.5 meters using the visual angle formula (1), ensuring reliable eye-tracking within the room-scale VR experience.

$$ObjectDistance = \frac{ObjectSize}{2 \cdot \tan\left(\frac{VisualAngle}{2}\right)} \quad (1)$$

3 RESULTS

3.1 Co-Presence

A two-way ANCOVA model was used to assess the impact of facial animation and expression on co-presence while controlling for baseline paranoia. Simple main effects analysis showed facial animation (group difference = 1.315, 95% CI = [0.001, 2.195], $F(1, 117) = 3.820$, $p = 0.053$, $\eta_p^2 = 0.032$) but not positive expression (group difference = 0.438, 95% CI = [-1.261, 0.967], $F(1, 117) = 0.094$, $p = 0.760$, $\eta_p^2 < 0.001$) led to a significant reduction in co-presence. There was no significant interaction between animation and friendly

¹Iclone LiveFace plugin: <https://mocap.reallusion.com/iclone-motion-live-mocap/iphone-live-face.html>

emotion ($F(1, 117) = 2.589$, $p = 0.887$, $\eta_p^2 < 0.001$). The ad-hoc test suggests there was no significant pair-wise difference.

3.2 Visual Attention

We excluded datasets from the eye-tracking analysis if more than 15% of the raw data was missing [5]. This resulted in the exclusion of five out of 122 datasets.

Analysing 117 participants during a 3-minute lift ride showed an average total fixation duration of 112.59 seconds ($SD = 28.42$) and 1.86 seconds ($SD = 1.84$) per fixation. Approximately 31.95% of the observed time was devoted to fixations on virtual humans ($SD = 24.68\%$). The average duration for these fixations was 5.87 seconds ($SD = 5.84$). The most common initial fixation targets (see Fig. 2) were the man directly facing the entry (27.4%), the exit (17.9%) and lift floor (17.9%).

A two-way ANOVA model was used to look at the visual attention devoted to virtual humans. There was a significant interaction between animation and positive expressions ($p = 0.047$, $\eta_p^2 = 0.034$) in the visual attention on virtual humans. Positive expressions ($p\text{-adj} = 0.046$) significantly decreased the visual attention to virtual humans when their faces were static. Tukey's HSD test for multiple comparisons showed there was a statistically significant difference between the *static neutral* and *static positive* group ($p\text{-adj} = 0.046$).

4 DISCUSSION & CONCLUSION

Our findings indicate that facial programming affects the attention and perception of virtual humans in individuals vulnerable to paranoia. Both animation and expressions could shape people's awareness and behavioural responses to these virtual humans.

Further analysis of people's paranoid appraisals and social thoughts towards virtual humans will provide deeper insights for interpreting the behavioural results. This will also help guide the future design of virtual humans in VR mental health applications.

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